

Venous Thromboembolism (VTE) Prophylaxis

Policy Number	POL-CLN-044	Owner / Department	Medical Staff / Pharmacy
Effective Date	2023-07-11	Version	2.1
Last Revised	2026-03-27	Next Review	2028-03-27
Approved By	Patient Safety & Quality Committee	Applies To	All licensed providers, pharmacists, and nurses caring for admitted adult medical and surgical patients

1. Purpose

To reduce hospital-associated venous thromboembolism (deep vein thrombosis and pulmonary embolism) through structured risk assessment and appropriate pharmacologic and/or mechanical prophylaxis, consistent with CHEST/ACCP antithrombotic guidelines.

2. Scope

Applies to every admitted adult patient. A VTE risk assessment and a bleeding-risk assessment are completed on admission, on transfer between levels of care, and at least every 24 hours thereafter or when clinical status changes.

3. Risk Assessment

Use the **Padua Prediction Score** for medical patients and the **Caprini Risk Assessment Model** for surgical patients. A Padua score ≥ 4 indicates high VTE risk; Caprini risk rises in steps with the cumulative score. Always pair the VTE risk score with a bleeding-risk assessment before prescribing pharmacologic prophylaxis.

Model / score	Risk tier	Recommended prophylaxis
Padua < 4 / Caprini 0–2	Low	Early ambulation; reassess daily.
Caprini 3–4	Moderate	Pharmacologic or mechanical prophylaxis per bleeding risk.
Padua ≥ 4 / Caprini ≥ 5	High	Pharmacologic prophylaxis unless contraindicated; add mechanical if high bleeding risk.

4. Prophylaxis Options

- **Pharmacologic:** enoxaparin 40 mg subcutaneously once daily (adjust for weight and renal function), or unfractionated heparin 5,000 units subcutaneously every 8–12 hours.
- **Mechanical:** intermittent pneumatic compression devices and/or graduated compression stockings when pharmacologic prophylaxis is contraindicated.
- Reassess renal function, platelet count (for heparin-induced thrombocytopenia), and bleeding daily.

IMPORTANT — Do not start or continue pharmacologic prophylaxis in a patient with active major bleeding, platelets $< 50 \times 10^9/L$, or another hard contraindication; use mechanical prophylaxis instead and document the reason. Every high-risk patient must have prophylaxis ordered or an explicit, documented reason for omission within 24 hours of admission.

5. Monitoring and Reassessment

- Reassess VTE and bleeding risk at least every 24 hours and at every transition of care.
- Investigate new unilateral limb swelling, pleuritic chest pain, dyspnea, or hypoxia for VTE promptly.
- Provide VTE prophylaxis education and address at-discharge extended prophylaxis where indicated.

6. Severity & Reporting

Report prophylaxis omissions and hospital-associated VTE per POL-RM-013.

Severity	Example	Notify / timeframe
SEV-1	Fatal or permanently disabling hospital-associated PE/DVT after prophylaxis was indicated but not ordered (sentinel-level event).	Attending, Administrator-on-Call, and Risk Management within 1 hour.
SEV-2	Hospital-associated VTE with major temporary harm, or bleeding from inappropriate prophylaxis.	Attending and Risk Management within 4 hours.
SEV-3	Missed or late risk assessment with no patient harm.	Unit manager within 24 hours via event report.
Near Miss	Omitted order caught by pharmacy/nursing and corrected before any harm.	Event report within 24 hours.

7. References

- CHEST / ACCP, Antithrombotic Therapy for VTE Disease and Prevention of VTE in Nonsurgical and Surgical Patients.
- Padua Prediction Score (medical) and Caprini Risk Assessment Model (surgical).
- Related: POL-RM-013.